



Alberta Public Health Disease Management Guidelines

Yersiniosis



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Health and Wellness Promotion Branch

Public Health and Compliance Branch

Alberta Health

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Case Definition

Confirmed Case

Laboratory confirmation of infection with or without clinical illness:^(A)

- Isolation of *Yersinia spp.* (except pestis) from an appropriate clinical specimen (e.g., stool)^(B) or food

Probable Case

Clinical illness^(A) with one or more of the following:

- Positive *Yersinia spp.* (except pestis) serology by agglutination test or ELISA

OR

- Detection of *Yersinia spp.* (except pestis) nucleic acid (e.g., PCR) in an appropriate clinical specimen (e.g., stool)

OR

- Epidemiological-linkage to a confirmed case

^(A) Clinical illness typically manifests as acute febrile diarrhea, enterocolitis (fever, diarrhea $\pm$ blood, severe abdominal pain), or acute mesenteric lymphadenitis mimicking appendicitis. Infection can be complicated by erythema nodosum, post-infectious arthritis, and systemic infection.

^(B) Refer to the [Public Health Laboratory \(ProvLab\) Guide to Services](#) for current specimen collection and submission information.

Reporting Requirements

Physicians, Health Practitioners and Others

Physicians, health practitioners and others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed and probable cases in the prescribed form by mail, fax or electronic transfer within 48 hours (two business days).

Laboratories

All laboratories shall report all positive laboratory results by mail, fax or electronic transfer within 48 hours (two business days) to the MOH (or designate) of the zone and the Chief Medical Officer of Health (CMOH) (or designate).

Alberta Health Services and First Nations and Inuit Health Branch

- The MOH (or designate) of the zone where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed and probable cases to the CMOH (or designate) within two weeks of notification and the final NDR (amendment) within four weeks of notification.
- The MOH (or designate) shall report all outbreaks to CMOH (or designate) via the [Alberta Outbreak Report Form \(AORF\)](#) as soon as possible using existing processes (e.g., CDOM or fax).
- For out-of-province and out-of-country reports, the following information should be forwarded to the CMOH (or designate) by phone, fax or electronic transfer within 48 hours (two business days):
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report, and
 - other relevant clinical/epidemiological information.

Epidemiology

Etiology

Yersiniosis is gram negative bacilli and a member of the family Enterobacteriaceae.^(1,2) There are two types of *Yersinia* which cause gastrointestinal illness: *Y. pseudotuberculosis* and *Y. enterocolitica*. *Yersinia pseudotuberculosis* is less likely to cause infection than *Y. enterocolitica*, the most common isolate in Canada.

There are other *Yersinia* species (e.g., *Y. intermedia*, *Y. frederiksenii*, *Y. kristensenii*); however, the pathogenic potential of these species is unclear.

Yersinia are susceptible to many disinfectants as well as moist and dry heat. Outside of the host, *Yersinia* can survive for up to 20 days in water and 540 days in soil.

Clinical Presentation

Yersiniosis usually presents as acute febrile diarrhea with abdominal pain. Enterocolitis occurs in approximately two-thirds of reported cases of *Y. enterocolitica*. In young children it presents as fever and diarrhea, in older children/adults as pseudoappendicitis syndrome (fever, right lower quadrant tenderness, abdominal pain).⁽³⁾

Individuals with *Y. pseudotuberculosis* can present with fever, scarlatinaform rash and abdominal symptoms, mimicking Kawasaki disease in children.⁽³⁾

Complications such as reactive arthritis and erythema nodosum can last up to six months.^(3,4) Other more rare complications include meningitis, endophthalmitis, conjunctivitis, myocarditis, pneumonia, pulmonary abscess, hepatitis, cholangitis, peritonitis, glomerulonephritis, urethritis, cellulitis, haemolytic anaemia, thyroiditis, pharyngitis and septicemia.

The mortality rate due to *Y. enterocolitica* septicemia in immunocompromised individuals, if untreated, can be as high as 50%.⁽²⁾

Diagnosis

Yersinia can be recovered from feces, throat swabs, mesenteric lymph nodes, peritoneal fluid, blood, vomitus, and contaminated food.⁽⁴⁾ Stool cultures are generally positive during the first two weeks of illness; it is important that laboratories be notified that *Yersinia* infection is suspected, as identification from stool specimens requires specific techniques.

Serology may be utilized as there can be difficulties in isolating *Yersinia* from feces. ELISA and agglutination tests may be ordered. The antibodies (against the O antigen) appear soon after the onset of illness and wane over two to six months. Paired sera specimens taken two weeks apart indicating a rise in agglutinating antibodies can support the diagnosis.

Treatment

Yersiniosis does not usually require treatment; however, antibiotics may be used to shorten the duration of symptoms and treat more severe, invasive infections.⁽⁵⁾

Reservoir

Yersinia enterocolitica is found mainly in swine; asymptomatic pharyngeal carriage is common.⁽⁴⁾ *Yersinia pseudotuberculosis* is a widespread zoonotic disease of birds and mammals, particularly rodents or small mammals.

Humans are incidental hosts of *Yersinia* infection.

Transmission

Transmission of *Yersinia* can occur by eating contaminated food, drinking contaminated milk or untreated water, person-to-person contact with someone or by contact with infected animals or their feces.⁽⁵⁾ Rarely, transmission may occur through contaminated blood via transfusion.

Incubation Period

The incubation period is most often three to seven days, generally less than 10 days.⁽⁴⁾

Period of Communicability

Fecal shedding occurs as long as symptoms persist, usually two to three weeks.^(3,4) If untreated, individuals may shed for two to three months. Prolonged asymptomatic carriage can occur in children and adults.

Host Susceptibility

Universal susceptibility. The disease is more common in children and young adults, and individuals who are immunocompromised are at increased risk of severe disease or death.⁽²⁾

Incidence

Although not a nationally notifiable disease, yersiniosis has been notifiable in Alberta since 1985.⁽⁶⁾ Between 2014 and 2017, over 200 cases were reported in the province. The majority (30%) were acquired via contact with contaminated food, followed by travellers outside of Alberta (22%).

Refer to the [Interactive Health Data Application \(IHDA\)](#) for more information.

Public Health Management

Key Investigation

- Confirm that the case meets the case definition.
- Obtain a history of illness including the date of onset, signs and symptoms.
- Identify any underlying medical conditions that may increase host susceptibility.
- Determine the occupation of the case and if in a **sensitive situation or occupation (SSO)** that poses a higher risk of transmission to others. See Table 1 for a definition of SSO.
- Determine the possible source of infection taking into consideration the incubation period^(C), reservoir, and mode of transmission. Assessment may include determining, identifying or obtaining:
 - a detailed food history including recent consumption of potentially contaminated food (especially poultry, beef, pork), contaminated water, or unpasteurized milk,
 - possible sources of cross contamination (e.g., cutting boards),
 - possible exposures to domestic or wild animals or fowl, including identifying recent illness in pets or acquisition of a puppy, kitten, etc. into the household,
 - occupational exposure (e.g., animal or meat handling),
 - recent travel or immigration,
 - residing in areas with poor sanitation, including improper water treatment and sewage disposal either in Canada or abroad, and/or
 - high-risk sexual practices, especially contact with feces,
- Obtain implicated food samples, if possible.⁽⁴⁾
- Assess for history of similar symptoms in other members of the household.
- Identify contacts that may have had significant risk of direct or indirect exposure to the feces of the case, especially those in SSO (refer to Table 1). Contacts include:
 - persons living in the household,
 - children and child care workers at a child care facility (daycare, dayhome, or other child care site), and
 - individuals exposed to the same source (if identified).

Table 1: Sensitive Situations or Occupations (SSO)

SSO	Activities
Food handler	• Touches unwrapped food to be consumed, <u>and/or</u> • Handles equipment or utensils that touch unwrapped food to be consumed.*
Health care, child care or other staff	• Has contact through serving food to <u>highly susceptible</u> patients or persons in whom an intestinal infection would have particularly serious consequences. • Provides direct patient care and is involved in the care of young children, elderly or dependent persons.
Child attending a child care facility or similar facilities	• Is diapered or unable to implement good standards of personal hygiene.
Any individual (older child or adult)	• Is unable to implement good standards of personal hygiene (e.g., with disabilities/challenges that may impact ability to perform good hand hygiene and is involved in an activity that may promote disease transmission.

* NOTE: Generally, food handlers who do not touch food, equipment or utensils in this way are not considered to pose a transmission risk; however, circumstances for each case should be assessed on an individual basis.

^(C) The incubation period of yersiniosis is usually 3–7 days; however, individual cases can vary requiring flexibility when determining where and how the disease was likely acquired.

Management of a Case

- All cases should be advised of the following:
 - how the disease is transmitted, appropriate personal hygiene, routine infection prevention and control practices, and contact precautions,
 - to avoid food preparation until symptoms have resolved, and
 - to practice safer sex and avoid sexual practices that facilitate fecal-oral transmission.
- Contact precautions should be used in health care settings where children or adults have poor hygiene or incontinence that cannot be contained.
- Refer to Table 2 for case exclusion criteria. For a summary of all enteric exclusions refer to [Enteric Transmission Risk Assessment and Exclusion Table](#).

Table 2: Case Exclusion

Cases	Category	Exclusion Criteria
Symptomatic	SSO	• The MOH may by order exclude the case until 48 hours after appropriate antibiotic treatment has been completed and stools have returned to normal or the MOH is satisfied that the case is no longer infectious. – The case must be symptom free for 48 hours after stopping any antidiarrheal medication (if taken). – Lifting of exclusions is not conditional upon submission of stool specimens to demonstrate clearance of the organism, – Specimens may still be submitted on a case-by-case basis in consultation with the MOH. • If possible, consideration may be given to temporary redeployment away from activities that involve increased risk of transmission.
Asymptomatic	SSO	• Generally not required unless otherwise recommended by the MOH.
Symptomatic	Non-SSO	• No exclusion required; however, all cases of gastroenteritis or enteritis should be regarded as potentially infectious and should remain home from work, school or daycare until 48 hours after diarrhea has stopped.
Asymptomatic	Non-SSO	• No exclusion required.

Management of Contacts

- Contacts should be instructed in disease transmission, appropriate personal hygiene, routine practices, and contact precautions.
- Symptomatic contacts should be assessed by a physician. Contacts with clinical illness should be managed and treated as cases.
- Refer to Table 3 for contact exclusion criteria.

Table 3: Contact Exclusion

Contacts	Category	Exclusion Criteria
Symptomatic	SSO	• Same as a case. • Ensure the contact is assessed by their physician.
Symptomatic	Non-SSO	• No exclusion required however, all cases of gastroenteritis or enteritis should be regarded as potentially infectious and should remain home from work, school or daycare until 48 hours after diarrhea has stopped. • Refer to their physician for assessment and testing, if required.
Asymptomatic	All	• No exclusion – contacts should monitor themselves for gastrointestinal symptoms, maintain good hand hygiene and food handling practices and seek medical attention if symptoms develop.

Preventive Measures

- Educate the **public** about the following:
 - personal hygiene, especially the sanitary disposal of items containing feces,
 - careful hand washing after defecation and sexual contact, and before preparing or eating food,
 - washing cutting boards, counter tops and utensils with soap and water after contact with raw meat
 - cooking meats thoroughly,
 - washing cutting boards, counter tops and utensils with soap and water after contact with raw meat (and other foods of animal origin);
 - the risk of keeping meat in the refrigerator for long periods of time before consumption, as *Yersinia* is able to multiply under refrigeration and microaerophilic conditions;
 - the dangers of consuming raw or undercooked meats, unpasteurized milk, and contaminated water;
 - cooking pork and other meats thoroughly;
 - methods of personal protective measures, in particular the correct and consistent use of condoms and risk of sexual practices that permit fecal-oral contact;
 - accessing and drinking safe water supplies; and
 - testing private water supplies for the presence of bacterial/parasitic contamination, if suspected.
- Educate **food handlers** about:
 - proper food and equipment handling, preparation, and hygiene, especially in avoiding cross-contamination from raw meat products; and
 - thorough hand washing before and after meal preparation.
- Educate **campers, backpackers, and others** to:
 - avoid drinking water directly from streams; and
 - boil water for at least one minute before it is used for drinking, food preparation, and oral hygiene.

Appendix 1: Revision History

Revision Date	Document Section	Description of Revision
November 2021	General	- Updated Template - Diagnosis and Treatment section moved to Epidemiology - Added reference and link to the Enteric Transmission Risk Assessment and Exclusion Table
	Reporting	Added a bullet on outbreak reporting
	Management of Case and Contacts	- Specified where terminology is SSO and included information in tables - Updated Exclusions to align with decisions agreed to in Enteric Transmission Risk Assessment and Exclusion Table - Re-worded to align with other enteric diseases
	Preventive Measures	Reworded to align with other enteric diseases

References

1. Government of Canada. Pathogen Safety Data Sheets: Infectious Substances – *Yersinia pseudotuberculosis* [Internet]. 2011. Available from: www.canada.ca/en/public-health/services/laboratory-biosafety-biosecurity/pathogen-safety-data-sheets-risk-assessment/yersinia-pseudotuberculosis.html
2. Government of Canada. Pathogen Safety Data Sheets: Infectious Substances – *Yersinia enterocolitica* [Internet]. 2011. Available from: www.canada.ca/en/public-health/services/laboratory-biosafety-biosecurity/pathogen-safety-data-sheets-risk-assessment/yersinia-enterocolitica.html
3. American Academy of Pediatrics. *Yersinia enterocolitica* and *Yersinia pseudotuberculosis* Infections. In: Kimberlin D, Brady M, Jackson M, Long S, editors. Red Book: 2015 Report of the Committee on Infectious Diseases. 30th ed. Elk Grove Village, IL: American Academy of Pediatrics; 2015. p. 868–70.
4. Heymann D, editor. Control of Communicable Diseases Manual. 20th ed. Washington, D.C.: American Public Health Association; 2015.
5. Centers for Disease Control and Prevention (CDC). Questions and Answers | *Yersinia* [Internet]. 2016. Available from: www.cdc.gov/yersinia/faq.html
6. Alberta Health. Communicable Disease Reporting System (CDRS). Edmonton, AB; 2019.